A Finite Black-Hole Information Ledger for Observer Patch Holography

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Abstract

A finite observer model separates unitary system–environment evolution from information discarded by a public record channel. For a supplied projective partition, pinching is an exact average of independent sign unitaries and its entropy gain equals the support-aware relative entropy to the pinched state. We state the additional access-algebra, state and geometric assumptions needed to interpret this identity as a black-hole information ledger. On a supplied three-dimensional layered population, crossing reads obey an area law with coefficient $(\pi/4)n^2a^4$. A declared nat per crossing fixes an area calibration, without deriving horizon entropy from the native record channel. The native observer provides finite unitary dilations and quantitative recovery limits, but no horizon, Hawking flux or evaporation dynamics. These distinctions separate the finite information identities from their conditional gravitational interpretation.

1 Unitary evolution and public records

For a finite density matrix ρ, distinguish evolution of the full system and record environment from the channel obtained by discarding part of that environment. Within the continuous star-automorphism class, the finite Stone theorem gives blockwise unitary evolution. A public record need not preserve quantum information: for a supplied partition $I = \sum_{j=1}^{k} P_j$, its pinching is

$$\mathcal{E}(\rho) = \sum_j P_j \rho P_j = 2^{-k} \sum_{\epsilon \in \{\pm 1\}^k} U_\epsilon \rho U_\epsilon^*, \qquad U_\epsilon = \sum_j \epsilon_j P_j.$$

The native record law is a distinct input. Its executed member is not idempotent, so it is not this pinching [9]. Keeping the complete native classical tape and final load register recovers all initial load populations, but not an arbitrary unknown quantum state. Its optimal entanglement fidelity is $D^{-2} \sum_h w_h \operatorname{rank}(P_h)^2$, equal to 1301/2028 for the full-carrier two-event C observer and 3329/6084 for E [8]. A black-hole interpretation requires an actual access algebra, a physical state and stress, and horizon dynamics on the same model.

2 Machine-checked inputs

Each input below is a machine-checked result in the repository, cited by module. The first two are standard finite-dimensional facts: the first is the Skolem–Noether theorem combined with Stone's

theorem on each Wedderburn block, and the second is character orthogonality on the group of diagonal signs. They enter as checked inputs, not as new results. Their joint consequence and the precise conjectures are stated in Section 3.

Theorem 2.1 (Finite Stone packet)**.** *Every continuous one-parameter group of star automorphisms of a unital, finite-dimensional private star subalgebra containing the ambient identity acts by blockwise unitary conjugation. On each Wedderburn block it is generated by a fixed self-adjoint Hamiltonian, unique up to a real scalar.*

Theorem 2.2 (Publicization is a uniform mixture of sign unitaries)**.** *For every supplied k-block projective partition, the pinching map is the uniform average of the 2^k diagonal sign unitaries (Lean module `EventAlgebra.RecordMajorization` [13]); averaging only the two global signs does not pinch, so the independence of the signs is load-bearing.*

Theorem 2.3 (Exact data processing and second law)**.** *Every stochastic kernel preserving a faithful stationary reference contracts Kullback–Leibler divergence to that reference (`kl_push_le`), the conditional heat bath obeys the second law (`heatBath_secondLaw`). For finite real-valued distributions and kernels, the integral fluctuation identity holds under its positivity, normalization and stationary-reference hypotheses; the pointwise Crooks identity additionally assumes detailed balance. The proofs are in the Lean modules `Thermodynamics.FiniteConditionalRepair` and `Thermodynamics.FluctuationTheorems` [13].*

Theorem 2.4 (Support boundary for entropy architecture)**.** *The totalized continuous-functional-calculus logarithm sends every projection to zero, so the raw finite trace formula for relative entropy assigns the value zero to an orthogonal pair of density projections whose support inclusion fails (`EventAlgebra.SpectralEntropyBoundary` [13]). A support-aware extended-value layer is therefore mandatory, not stylistic.*

Theorem 2.5 (Conditional rich-fibre regional packet)**.** *A declared adapter for four pairwise-disjoint observer windows constructs noncommutative regional blocks, each with a nonunital two-by-two designated corner, elementwise commutation across blocks, four-window coverage and unique restriction gluing. The source supplies split-fibre labels; the matrix units, state, restrictions and repair are adapter data. The corner is not a tensor factor or a tensor-split theorem (`QFT.RichFibreWitness`, `QFT.RichFibreRegionalNet`) [13].*

Theorem 2.6 (Finite KMS identity, degenerate inheritance boundary)**.** *The finite algebraic KMS identity holds for Gibbs states of support-graded flows, and the truncation-8 nets inherit KMS structure only degenerately (`QFT.StructuralInheritance` [13]). The nondegenerate continuation on the rich-fibre net is Conjecture 3.6.*

Because the native full configuration reference has a tracial limit at fixed modular parameter, its limiting modular automorphism is the identity, and it supplies no nondegenerate KMS state of this kind [11].

3 Finite conjectures

Throughout, ρ is a finite density matrix on the ambient algebra, $\mathcal{E}$ the pinching of a supplied projective partition, S the support-aware von Neumann entropy, and $D(\cdot\,\|\,\cdot)$ the support-aware relative entropy. The statements of this section add finite hypotheses to the preceding theorems and do not follow from them. Two statements of this section are known theorems that the library has not formalized; they are labeled accordingly.

Known theorem 3.1 (Support-aware entropy core)**.** *For the pinching $\mathcal{E}$ of a projective partition,*

$$S(\mathcal{E}(\rho)) - S(\rho) = D(\rho \,\|\, \mathcal{E}(\rho)) \geq 0,$$

with equality exactly at the public fixed states, where ($\mathcal{E}(\rho) = \rho$), and the sign-unitary mixture is entropy-nondecreasing. This core is the subset of the spectral-information chain required by the ledger; the more general majorization, maximum-entropy, and H-theorem scope is not used here.

Standard proof. The pinching is the uniform average of 2^k sign conjugations, so $\mathcal{E}(\rho) \geq 2^{-k}\rho$ and the support of ρ lies in that of $\mathcal{E}(\rho)$; the relative entropy is finite. The operator $\log \mathcal{E}(\rho)$ is block diagonal, and $\mathcal{E}$ is a trace-preserving conditional expectation onto block-diagonal operators, so $\operatorname{Tr} \rho \log \mathcal{E}(\rho) = \operatorname{Tr} \mathcal{E}(\rho) \log \mathcal{E}(\rho)$. Hence $S(\mathcal{E}(\rho)) - S(\rho) = \operatorname{Tr} \rho \log \rho - \operatorname{Tr} \rho \log \mathcal{E}(\rho) = D(\rho \,\|\, \mathcal{E}(\rho))$, which is nonnegative with equality exactly when $\rho = \mathcal{E}(\rho)$ by Klein's inequality. The library does not formalize this support-aware extended-value identity; the support boundary theorem explains why the raw logarithm formula is insufficient. □

Known theorem 3.2 (Finite no-loss statement)**.** *For a specified finite unitary dilation retaining its record environment, joint entropy is preserved. Co-evolving two states under the same unitary also preserves their support-aware relative entropy and trace distance.*

This is unitary invariance. Its physical premise is the specified dilation and access to its environment; it is not a statement about arbitrary coordinate-dependent divergences or channels that discard the records.

The record-discarded native repair channel is not of this kind: it strictly attenuates a coherent mode, which rules out unitary conjugation on the carrier, and under independent seam attempts the unread binary fixed-number state converges to its tracial entropy maximum. Each finite native observation history has an exact unitary representation once its record environment is kept, with a dimensionless Hermitian action A_0 whose exponential $\exp(-iA_0)$ is that unitary representation [5, 6, 7].

Conjecture 3.3 (Ledger identity and relocation)**.** *For the specified record channel, the public entropy gain equals $D(\rho \,\|\, \mathcal{E}(\rho))$; the quantity is nonnegative, vanishes exactly at the public fixed states, and is carried entirely by the off-block coherences that no access cut of the specified net exposes. Information is relocated, never destroyed, and the relocation is exactly quantified.*

The equality and its sign follow from Known theorem 3.1. Inaccessibility to every cut of the specified net is an additional assumption.

Conjecture 3.4 (Generalized-second-law-shaped monotone)**.** *The composition of the cap first law with Kullback–Leibler contraction yields one exact monotone combining record entropy and the capacity ledger under repair, with finite constants and a negative control exhibiting failure of the monotone when unitality of the record channel is dropped.*

A classical sector-level instance follows from the joint counting identity $S_{\mathrm{count}}(p, d) = H(p) + \sum_i p_i \log d_i = \log M - D(p \,\|\, d/M)$: a repair kernel on sectors that preserves d/M contracts the divergence by `kl_push_le`, so S_{count} does not decrease, while a reset kernel with stationary law $(1/4, 3/4)$ on two equal sectors does not preserve d/M and lowers S_{count} from its maximum [10]. The conjecture concerns the quantum record channel and the cap first law.

Conjecture 3.5 (Complementarity receipts)**.** *There is a composition of the generic operational access-cut interface, the supplied rich-fibre data, and a justified operator/local-channel adapter that*

forms a two-observer ledger in which each accessible algebra determines its own record entropy, the declared block-commutation theorem supplies an algebraic locality receipt, and a glued top observable is uniquely reconstructed within the declared restriction system. No physical no-signalling or refinement-natural interpretation follows before that attachment.

Conjecture 3.6 (Nondegenerate finite KMS receipt)**.** *The modular data of a restricted state on one genuinely noncommutative rich-fibre region satisfies the finite KMS identity at a capacity-linked inverse temperature, nondegenerately. This receipt supplies a finite KMS input but does not establish structural QFT thermality.*

Conjecture 3.7 (Finite Page-crossover inequality)**.** *For a bipartition of the specified finite net, the exact 2^k sign-unitary average obeys a finite inequality that locates the crossover of accessible-record information without Haar integration.*

Remark 3.8 (Conditional finite implication). Known theorems 3.1 and 3.2 with Conjectures 3.3–3.6 would imply simultaneous ambient unitarity and a thermal public record, with the inaccessible information equal to the computable functional $D(\rho \,\|\, \mathcal{E}(\rho))$ and localized in observer-inaccessible coherences. The implication supplies no physical black-hole attachment. A source-selected order embedding with calibrated density (a causal-set faithful embedding), manifoldlikeness, and a calibrated identification of native event clocks with proper time are additional hypotheses. Any smooth Einstein reading also requires same-family tensor-curvature convergence or the independent continuum small-ball/null-balance identification.

4 Crossing-read area law on the source family

The layered read law of the conservative source family supplies a finite count that scales with area. Fix a population of constant density n, sites per unit volume, in a bounded convex window $\Omega \subset \mathbb{R}^3$, a read radius a, and the complete-neighbour read law: an event at layer $j+1$ reads every event at layer j within distance a, the same site included. For a plane section $\Sigma = \Omega \cap \{x_1 = P\}$, a crossing read of one layer is an unordered site pair $\{s, t\}$ with $s \neq t$, $\|t - s\| \leq a$, and $s_1 < P < t_1$. Each such pair is realized in every layer by one read in each direction across Σ, from the lower layer to the upper layer; the ordered read count is therefore twice the unordered count, and the unordered count per layer is the primary quantity here. A same-site read never crosses a plane, and a site with $s_1 = P$ belongs to neither side.

Proposition 4.1 (Crossing reads per unit area)**.** *Let $S_k \subset \Omega$ be finite populations with a partition of Ω into cells of common volume $1/n_k$, each cell contained in the ball of radius H_k about its site, with read radii $a_k \to 0$ and $H_k/a_k \to 0$. Let $\Sigma = \Omega \cap \{x_1 = P\}$ have positive area $|\Sigma|$. The number X_k of unordered crossing reads of Σ per layer satisfies*

$$X_k = n_k^2 \, |\Sigma| \int_0^{a_k} \frac{\pi (a_k - u)^2 (2a_k + u)}{3} \, \mathrm{d}u \, (1 + o(1)) = \frac{\pi}{4} \, n_k^2 a_k^4 \, |\Sigma| \, (1 + o(1)). \tag{1}$$

The relative error is bounded by a constant multiple of $H_k/a_k + a_k$, the constant depending on Ω and Σ.

Proof. Write $u(s) = P - s_1$ for the depth of a site below Σ and $C(s) = \{t \in \Omega : \, t_1 > P, \, \|t - s\| \leq a_k\}$, so that $X_k = \sum_{0 < u(s) \leq a_k} |S_k \cap C(s)|$. The cell of a site t lies in $B(t, H_k)$; a cell that meets both $C(s)$ and its complement lies in the $2H_k$-collar of $\partial C(s)$. Hence $|S_k \cap C(s)|$ differs from $n_k \operatorname{vol} C(s)$ by at most n_k times the volume of that collar. The boundary $\partial C(s)$ consists of a spherical cap,

a planar disc and pieces of $\partial\Omega$, of total area $O(a_k^2)$, so the error is $O(n_k H_k a_k^2)$ per site. The slab $0 < u(s) \le a_k$ contains $O(n_k a_k |\Sigma|)$ sites, which gives a total error $O(n_k^2 H_k a_k^3 |\Sigma|)$, of relative order H_k/a_k.

The function $f(s) = n_k \operatorname{vol} C(s)$ on the slab is bounded by $\frac{2\pi}{3} n_k a_k^3$ and is Lipschitz with constant $O(n_k a_k^2)$: its depth derivative is $-\pi n_k (a_k^2 - u^2)$ and a lateral displacement changes $C(s)$ through the faces of Ω by a set of measure at most πa_k^2 per unit displacement. The same cell argument, applied to the slab, bounds $\left|\sum_{0<u(s)\le a_k} f(s) - n_k \int_{0<u\le a_k} f \,\mathrm{d}^3 s\right|$ by n_k times the Lipschitz constant times H_k times the slab volume, plus the contribution of the sites within H_k of the two slab faces; both are $O(n_k^2 H_k a_k^3 |\Sigma|)$, again of relative order H_k/a_k.

Finally, for s at lateral distance more than a_k from the relative boundary of Σ in the plane, $C(s)$ is the untruncated spherical cap of $B(s, a_k)$ beyond height u, with volume

$$\int_u^{a_k} \pi(a_k^2 - z^2)\,\mathrm{d}z = \frac{\pi(a_k - u)^2(2a_k + u)}{3}.$$

The truncated sites occupy a lateral collar of area $O(a_k)$ relative to $|\Sigma|$, so the integral over the slab equals $|\Sigma| \int_0^{a_k} \pi(a_k - u)^2(2a_k + u)/3 \,\mathrm{d}u$ up to a relative error $O(a_k)$. The depth integral is $\pi a_k^4/4$. □

Corollary 4.2 (Golden-family value)**.** *For the golden source population S_q of q^3 sites in $\Omega = [0, L]^3$, $L = 2/\sqrt{\varphi + 2}$, with $n_q = q^3/L^3$, $a_q = L/\sqrt{q}$ and cell assignment error $H_q = 2\sqrt{3}L/q$, one has $n_q^2 a_q^4 = q^4/L^2$, and the unordered crossing count per layer through a full cross-section $\Sigma = \Omega \cap \{x_i = P\}$ is*

$$X_q = \frac{\pi q^4}{4}(1 + O(q^{-1/2})). \tag{2}$$

For the mid-plane $P = L/2$ the continuum count with the caps truncated by the cube is

$$n_q^2 \int_{s\in\Omega,\, s_1<L/2} \operatorname{vol}\{t \in \Omega :\ t_1 > L/2,\ \|t - s\| \le a_q\}\,\mathrm{d}^3 s = q^4 \left(\frac{\pi}{4} - \frac{8}{15\sqrt{q}} + \frac{1}{12\,q}\right). \tag{3}$$

Proof. Here $H_q/a_q = 2\sqrt{3}\,q^{-1/2}$ and $a_q/L = q^{-1/2}$, so Proposition 4.1 gives (2). For (3) substitute $d = t - s$. For $0 < d_1 \le a_q \le L/2$ and $|d_2|, |d_3| \le L$, the set of $s \in \Omega$ with $s + d \in \Omega$ and $s_1 < L/2 < s_1 + d_1$ has volume $d_1(L - |d_2|)(L - |d_3|)$. Over the half ball $d_1 > 0$, $\|d\| \le a_q$, the moments are $\int d_1 = \pi a_q^4/4$, $\int d_1 |d_2| = 4a_q^5/15$ and $\int d_1 |d_2||d_3| = a_q^6/12$. Hence the count equals $n_q^2 a_q^4(\pi L^2/4 - 8La_q/15 + a_q^2/12)$, which is (3) after $n_q^2 a_q^4 = q^4/L^2$ and $a_q = L/\sqrt{q}$. □

Proposition 4.3 (Area law and the Planck-area dictionary)**.** *Declare that each crossing read carries one nat of cut entropy across Σ per layer. In the limit of Proposition 4.1 the cut entropy is the area law*

$$S_{\mathrm{cut}}(\Sigma) = \frac{\pi}{4}\, n^2 a^4\, |\Sigma|. \tag{4}$$

Identifying (4) with $|\Sigma|/(4\ell_P^2)$ fixes

$$n^2 a^4 = \frac{1}{\pi \ell_P^2}, \qquad \ell_P = \frac{1}{\sqrt{\pi}\, n\, a^2}, \tag{5}$$

and on the golden family $\ell_P = L/(\sqrt{\pi}\, q^2)$.

Proof. Equation (4) is (1) with one nat per read. Equating $(\pi/4)n^2a^4$ with $1/(4\ell_P^2)$ and solving for ℓ_P^2 gives (5); with $n^2a^4 = q^4/L^2$ this reads $\ell_P^2 = L^2/(\pi q^4)$. □

Remark 4.4 (What is not claimed). Equation (5) is a calibration of the read scale by the area coefficient, in the same way that the layer duration a/c is a calibration of the clock. No horizon is constructed: Σ is a plane section of the source window, with the same read law on both sides. The nat-per-read reading is a declared dictionary, without a derivation from the record channel of the finite ledger. No temperature, no Hawking flux and no physical value of ℓ_P follow; L is the side of the source window, and the family fixes only the ratio $\ell_P/L = 1/(\sqrt{\pi}\, q^2)$. Dou and Sorkin [4] count causal links across a horizon in a Poisson-sprinkled causal set and obtain an area law whose coefficient depends on the sprinkling; here the population is deterministic, the count is exact at every finite q, and the coefficient $\pi/4$ in (1) is the exact limit.

The depth integral in (1), the value $(\pi/4)n^2a^4$, the golden identity $n_q^2 a_q^4 = q^4/L^2$ with the cross-section value $\pi q^4/4$, and the dictionary (5) with its golden value $\ell_P^2 = L^2/(\pi q^4)$ are machine-checked in the Lean module `CrossingReadAreaLaw` (`capVolume_integral`, `crossingDensity_eq`, `golden_crossing_count`, `planck_area_of_area_law`, `golden_planck_length_sq`) with the standard axioms only. The limit statement of Proposition 4.1 is analytic and is proved above. The crossing-read count is a declared one-nat-per-read quantity on a supplied three-dimensional population. On the two-dimensional native source screens, regional reduced entropy carries no nonzero coefficient times the boundary-seam count, so this law is distinct from a native entropy-area law; a whole-screen reading as the boundary of a three-dimensional region needs its own region map.

Finite receipt. The exact crossing counts of the golden family at $q = 5, 8, 13, 21, 34, 55$ are recorded with an independent verifier. The population and the neighbour relation are rebuilt from the definitions, the relation agrees edge for edge with the causal-poset evidence package at every level, every side decision is exact in $\mathbb{Q}(\varphi)$, and no site lies on any of the five planes $x_1 = L/2$, $x_1 = L/4$, $x_1 = 3L/4$, $x_2 = L/2$, $x_3 = L/2$. Table 1 lists the mid-plane counts. The counts across $x_2 = L/2$ and $x_3 = L/2$ equal the $x_1 = L/2$ count at every level, as the product population is symmetric under coordinate permutations. At $q = 34$ the mid-plane count is 907 080 against the box-corrected value 930 602.0, ratio 0.975, and against the infinite-plane value 1 049 555.8, ratio 0.864; at $q = 55$ the count is 6 496 004 against 6 542 684.6, ratio 0.993, and against 7 186 884.1, ratio 0.904. The ratio to the box-corrected value is closer to one than the ratio to the infinite-plane value at every level; it lies within 0.13 of one at every level and within 0.026 of one for $q \geq 21$, while the ratio to the infinite-plane value rises from 0.658 to 0.904. The residual at fixed q is the finite population itself: the two sides of the mid-plane hold 84 700 and 81 675 sites at $q = 55$, and the mean number of neighbours of a site, 1459.3 at $q = 55$, compares with the cube mean 1461.1 of $n_q \operatorname{vol}(B(s, a_q) \cap \Omega)$ and with the untruncated ball value 1708.6.

q	count	$\pi q^4/4$	ratio	box-corrected	ratio
5	323	490.9	0.658	352.2	0.917
8	2 164	3 217.0	0.673	2 487.3	0.870
13	17 754	22 431.8	0.791	18 390.1	0.965
21	129 830	152 745.0	0.850	130 882.5	0.992
34	907 080	1 049 555.8	0.864	930 602.0	0.975
55	6 496 004	7 186 884.1	0.904	6 542 684.6	0.993

Table 1: Unordered crossing reads per layer across the mid-plane $x_1 = L/2$ of the golden family, the infinite-plane value $\pi q^4/4$, and the box-corrected value (3), with the ratios of the count to each.

Declared cut entropy and native regional entropy. The population, its read radius and the layer read law of the preceding section are specified inputs, and the cut entropy is a declared count on read links; the seam-count wording of that section refers to these read links, not to native repair seams. It is distinct from native regional entropy. On a source screen of $N = 240 \cdot 4^l$ ports, a complete refining base-face patch with $12 \cdot 4^l$ ports has $3 \cdot 2^l$ crossing seams; its stationary reduced entropy grows in proportion to 4^l, while the return reference's mutual information is at most $\log(12 \cdot 4^l + 1) + 2025/(64N)$, which per crossing seam tends to zero. Neither native regional quantity carries a nonzero universal coefficient times boundary-seam count on that two-dimensional source [10]. A reading of a whole screen's cell count as the area of a three-dimensional region's boundary is not excluded, and that region map is a premise. The Lean statements of the preceding section are in the module `Geometry.CrossingReadAreaLaw` [13].

5 Physical interpretation

The Page, Hayden–Preskill and island analyses concern specified random, holographic or gravitational systems [1, 2, 3]. The finite identities here supply no evaporating horizon or gravitational state. Their attachment requires a common causal geometry, calibrated clock, conserved stress and horizon-dependent access algebra. Neither a declared read count nor a finite thermal state proves Hawking radiation.

6 Conditional ringdown template

A separate transition template assumes positive integer record dimensions $d_{\text{before}} = k d_{\text{after}}$. Thus k divides the initial dimension and the black-hole entropy change is $-\log k$. Combining this assumption with the leading small-transition Kerr first law gives

$$x_k = \frac{\log k}{8\pi}, \qquad x = \frac{GM}{c^3 g(\chi)}(\omega - m\Omega_H), \qquad g(\chi) = \frac{2\sqrt{1-\chi^2}}{1+\sqrt{1-\chi^2}}.$$

For $M > 0$, $|\chi| < 1$, and denominator tooth $k_b \geq 2$, offset-subtracted ratios are $\log k_a / \log k_b$. Observed frequencies require the same detector-frame mass $M_{\text{det}} = (1+z)M_{\text{source}}$ in both frequency terms. At extremality $g = 0$: the teeth collapse to the rotation line and the rescaled ratios are undefined.

The declared linewidth-to-spacing ratio is $64\pi^2 p_0/[a g(\chi)^2 \log k]$; it is mass-independent and spin-dependent, with no controlled constant-Page-coefficient approximation near extremality. The factor $(k-1)/k$ is a within-line KMS response, not a probability over transitions. No source-derived transition rule, finite-step correction or event likelihood is supplied. The finite $2 \leq k \leq 12$ ladder excludes neither other integers nor the full family if rejected. A test requires a detector likelihood, nuisance model, normalized integer prior or finite submodel, selection and trials rule, greybody normalization, no-comb reference and numerical convergence criterion. Posterior samples alone do not supply that likelihood. The template and its exact algebraic controls are retained in the prediction register [12]; it establishes no black-hole observation.

7 Verification surface

The proved theorems cited in Section 2 carry `#print axioms` receipts with only the standard axioms, no admissions, and no native-evaluation shortcut, in the modules named inline. The

support-aware entropy statements use the general spectral-information definitions. Known theorems 3.1 and 3.2 with Conjectures 3.3–3.6 are not proved by those receipts.

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